

Education

Texas A&M University

Expected in May 2027

B.S.: Multidisciplinary Engineering Technology, Minor in Embedded Systems Integration

College Station, TX

GPA: 3.3

Skills

- **Robotics & Controls:** Kinematics, Closed-Loop PID Tuning, Motion Planning, State Estimation
- **Embedded Systems:** C/C++, Microcontrollers (STM32/Arduino), SBCs (Raspberry Pi), I2C/SPI/UART/PWM
- **Perception & Software:** Python, ROS/ROS 2, Computer Vision (OpenCV), Sensor Fusion (LiDAR, IMU, Cameras)
- **Electromechanical:** Actuators, Motor Drivers, Sensor Integration, Power Systems, CAD & Prototyping

Experience

Tex-Tube Steel

May 2026 to August 2026

Project Engineering Intern

Houston, TX

- Developed Python automation tools to digitize manufacturing workflows, reducing manual data entry and optimizing internal data flow.
- Automated engineering processes by integrating custom software solutions directly with enterprise SAP systems to increase process reliability.

SOMTECH Robotics

September 2024 to Current

Mechanical Lead / End Effector Lead

College Station, USA

- Directed the mechanical engineering team for the University Rover Challenge Mars rover, overseeing system architecture, CAD analysis, and mechanical integration.
- Led design and prototyping of an underactuated robotic gripper end-effector for specialized object manipulation.
- Collaborated with electrical and software teams to integrate multi-axis actuation, sensing networks, and control loops into a cohesive platform.

Texas A&M University at Qatar

May 2023 to July 2023

AI/ML Research Intern

Al Rayyan, Qatar

- Built, trained, and evaluated machine learning models using TensorFlow and Python to explore neural network convergence.
- Applied hyperparameter tuning and metric-driven evaluations to improve baseline model classification accuracy.

Selected Projects

SCUTTLE Search & Rescue Autonomous Mobile Robot

- **Kinematics & Control:** Calculated differential drive kinematics and tuned multi-loop PID controllers for trajectory tracking and motor actuation.
- **Perception & SLAM:** Interfaced LiDAR and RGB camera data to execute 2D SLAM mapping, dynamic obstacle avoidance, and target identification algorithms.
- **Navigation Architecture:** Implemented a navigation stack combining sensor fusion, global/local path planning, and finite-state machine decision logic.
- **Hardware/Software Integration:** Integrated high-torque DC motors, motor drivers, sensor power distribution networks, and Linux embedded hardware.

Vision-Guided Autonomous Mobile Robot (AMR)

- **Embedded Processing:** Built a real-time Linux/Python processing pipeline on Raspberry Pi for sensor sampling, state estimation, and motor command execution.
- **Closed-Loop Actuation:** Configured closed-loop PID velocity control using optical encoder feedback and hardware PWM signals to H-bridge motor drivers.
- **Vision & Inertial Tracking:** Fused IMU telemetry with OpenCV color-space detection (HSV filtering) for closed-loop trajectory correction.
- **Hardware Interfacing:** Designed power regulation circuits, L298N motor drivers, and I2C/SPI sensor buses.